

Trace metal storage in recent floodplain sediments along the Dill River, central Germany

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ARTICLE INFO

Article history:

Received 9 August 2014

Received in revised form 23 January 2015

Accepted 27 January 2015

Available online 7 February 2015

Keywords:

Trace metals

Recent floodplain sedimentation

Industrial/mined watersheds

Sediment contamination

Germany

ABSTRACT

Trace metals are stored in near-channel floodplain sediments along many rivers in industrial and mined watersheds of western European countries such as Germany. In this paper, I document the distribution of Cu, Pb, and Zn in 13 cores collected from near-channel floodplain sediments along a 25-km reach of the Dill River in central Germany. Mean concentrations of the three trace metals exceed background concentrations, an indication of anthropogenic enrichment along the study reach; many individual samples have concentrations several times the background levels and exceed standards for trace metals contained in the German Federal Soil Protection Act. Metal concentrations generally peak at depths of 0.2 to 0.7 m below the floodplain surface and are assumed to represent the period of maximum metal releases to the environment through upstream industrial activity, ore mining, or both. In eight of the cores, radiocarbon ages obtained below the depth of peak metal concentrations provide maximum dates for sediment accumulation of between 90 and 1700 years ago, with most ages clustering in the period of 90 to 300 years ago. The ages and depth to peak metal concentrations indicate 0.6 to 1.0 m of floodplain sedimentation over the last 150 to 300 years ($0.3\text{--}0.4\text{ cm y}^{-1}$), which exceed sedimentation amounts and rates found in surrounding watersheds. Surface sediments contain lower metal concentrations, suggesting deposition recently of cleaner sediments. Trace metals stored along the Dill River serve as a reminder that stores of contaminants exist in geomorphically sensitive locations of the fluvial system. These *legacy* sediments bear the chemical imprint of industrial or mining activities that no longer occur in the watershed.

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1. Introduction

Contamination with trace metals is widespread in numerous watersheds worldwide. The source of the metals varies but often can be linked to industrial activity, agriculture, mining, or a combination thereof. In areas such as western Europe, metals have been discharged to the environment for centuries (Matschullat et al., 1997; Brännvall et al., 1999; Renberg et al., 2001), although the maximum releases generally occurred between the mid-nineteenth century and early 1970s (Valette-Silver, 1993; Renberg et al., 2001; Middelkoop, 2002; De Vleeschouwer et al., 2007). The earliest source of trace metal contamination is mining, which in parts of Europe extends back more than 3000 years (Monna et al., 2000). Many studies have documented the long-term environmental impacts of metal storage in, and conveyance through, mined and industrial watersheds (Leigh, 1994, 1997; Miller, 1997; Macklin et al. 2006; Bábek et al., 2008; Lecce et al., 2008; Foulds et al., 2014; Lecce and Pavlowsky, 2014; Resongles et al., 2014). Trace metals impact the fluvial environment in many ways, including water quality, aquatic ecosystems, and the health of humans (Lecce and Pavlowsky,

2014). Although mining ceased in many contaminated watersheds decades earlier, debris generated by the mining process is stored as floodplain and channel sediments, serving as a source for trace metal pollution downstream. When metals are stored in geomorphically sensitive areas such as channel banks or near-channel positions, they may be readily remobilized and conveyed downstream by multiple or single events (James, 1989; Marcus, 1989; Leigh, 1997; Förstner, 2003; Krüger et al., 2005; Foulds et al., 2014).

In western Europe, the impact of past mining on floodplain and channel bank sediments has been well documented (Hudson-Edwards et al., 1999; Mascaro et al., 2001; Coulthard and Macklin, 2003; Hürkamp et al., 2009; Resongles et al., 2014). Trace metal contamination has been noted in floodplain sediments and channel banks tens of kilometers below historically mined reaches of watersheds. Frequently, metal concentrations decrease with distance from the former mines, a trend usually attributed to deposition of a majority of metals in close proximity to the mined areas and the addition of cleaner sediments from downstream tributaries. Highly contaminated channel banks and near-channel floodplain sediments have been shown to serve as a source for metal contamination during floods (Leigh, 1994; Schwartz et al., 2006; Ciszewski and Turner, 2009; Wyzga and Ciszewski, 2010; Foulds et al., 2014). Industrial activity can also serve as a source of trace metals, as has been documented in western Europe along rivers such as the Rhine

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(Dehner, 1994; Middelkoop, 2002; Wijnhoven et al., 2006) and Elbe (Förstner, 2003; Krüger and Gröngroft, 2003; Schulz-Zunkel et al., 2013; Zachmann et al., 2013).

Work that I have undertaken since the mid-1990s along the Lahn River, a predominantly agricultural watershed of ~5000 km² in west-central Germany, served as the impetus for the present study (Fig. 1). My previous research documents anthropogenic trace metal storage in floodplain sediments along a 75-km reach of the Lahn River, examining the geographic pattern of trace metal storage across the floodplain (Martin, 1997, 2000) and the change in near-channel storage of trace metals since the mid-1990s (Martin, 2009, 2012). Along the Lahn River downstream of its confluence with the Dill River in Wetzlar, trace metal concentrations increase (Martin, 2012). Although the Lahn River has no history of mining upstream of Wetzlar, mining operations were present in the Lahn River downstream from Wetzlar and in the upper Dill River from the 1830s until the last mine closed in 1983; the peak of mining activity in the Dill and lower Lahn River valleys was between 1870 and 1945 (Meyer, 2007). Historically, steel mills also operated in the upper part of the Dill River watershed (Fröhlich et al., 2008). Preliminary sampling of channel banks along the Dill River revealed Cu, Pb, and Zn concentrations above those found in similar geomorphic positions along the Lahn River (Martin, 2012). The research reported here focuses on trace metal storage in additional near-channel sites

along a 25-km reach of the Dill River and uses trace metal vertical trends and radiocarbon ages to document recent floodplain sedimentation.

The impact of historical mining activity on rivers and streams has been extensively documented in the UK (e.g., Hudson-Edwards et al., 1999; Coulthard and Macklin, 2003; Dennis et al., 2009; Foulds et al., 2014). In Germany, mining extends back many centuries, with some of the oldest documented activity dating to around 3500 Y BP in the Harz Mountains of central Germany (Matschullat et al., 1997; Monna et al., 2000). A significant expansion of mining occurred there in AD 980 with the establishment of the Rammelsberg mines (Matschullat et al., 1997; Brännvall et al., 1999; Bindler et al., 2011). The northern Vils River of eastern Bavaria, where mining began about 1000 Y BP, is another area of early historical mining in Germany (Hürkamp et al., 2009). Typically, mining released large quantities of trace metals to the environment that were subsequently conveyed downstream and stored in floodplains. Today, they provide an *archive* of past mining activity and the resultant metal contamination (Hürkamp et al., 2009). Vertical trends in trace metals, which have been used to document movement of metal-contaminated sediments through drainage basins, permit assessment of the potential for near-channel floodplain sediments to serve as future sources of downstream contamination (Gocht et al., 2001; Middelkoop, 2002; Lecce and Pavlowsky, 2014; Vallius, 2014). When age control on vertical metal trends is available, either

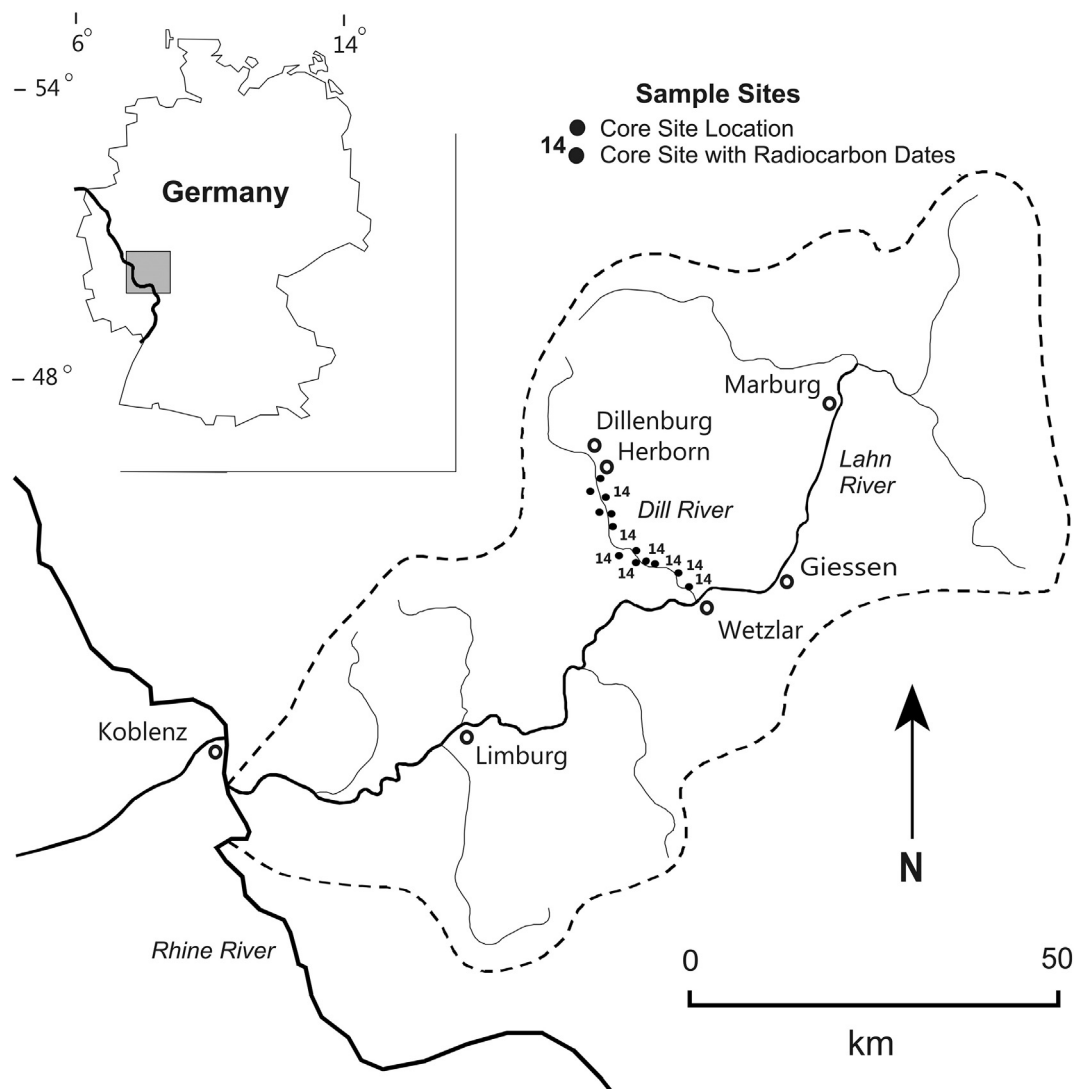


Fig. 1. Study reach of the Dill River between Herborn and Wetzlar. Sample core locations ($n = 13$) are shown with solid dots; sites with radiocarbon control are indicated with '14'.

through knowledge of the upstream industrial or mining history or through radiometric ages, calculation of recent overbank sedimentation rates is possible (Krüger et al., 2006; Lecce et al., 2008; Matys Grygar et al., 2011; Lecce and Pavlowsky, 2014; Resongles et al., 2014).

In central Germany, the late Holocene alluvial record has been compiled on a timescale of thousands of years (Lang and Nolte, 1999; Rittweger, 2000; Bos and Urz, 2003; Houben, 2003, 2012; Kalis et al., 2003; Wennrich et al., 2005; Hoffmann et al., 2007), but an understanding of alluvial sedimentation rates over shorter time-scales of hundreds of years is lacking. The present study provides new information on alluvial sedimentation over the last 300 years using trace metal trends and radiocarbon ages on stratigraphy. Within the broad goal of examining trace metal storage in near-channel floodplain sediments of the Dill River, the specific objectives of this research are to (i) document the extent of trace metal storage in near-channel floodplain sediments along a 25-km reach of the Dill River from Herborn to the mouth of the river at Wetzlar; (ii) examine the vertical trends in trace metal concentrations at the near-channel sites; (iii) using radiocarbon ages and vertical trace metal trends, estimate recent floodplain sedimentation rates; and (iv) compare trace metal concentrations in the Dill River valley with concentrations in the adjacent Lahn River valley upstream from the confluence of the two rivers.

2. Study area

The Dill River, with headwaters in the Lahn–Dill low mountain range of west-central Germany, drains an area of ca. 700 km² (Fig. 1). Along its length, the river drops from 700 to 150 m, flowing straight southward through the cities of Dillenburg and Herborn for a distance of nearly 60 km before emptying into the Lahn River at Wetzlar. Unlike the Lahn River, the Dill River has not been extensively channelized, resulting in higher flow velocity and stream power than along the Lahn River. Overbank flow generally occurs during the winter and early spring (December to April), a period of higher precipitation and potential snowmelt from upland areas (Breuer et al., 2009). The lithology of the Dill basin is highly variable, with Tertiary metabasalts and greenstone in the upper and lowermost parts of the study reach and Carboniferous siltstone and shale in the middle portion (Fröhlich et al., 2008). Slightly more than 50% of the watershed is forested (mainly uplands) and 20% is in pasture (lowlands and floodplain), with small amounts of arable land (7%) in the southern-most part of the study reach (Fröhlich et al., 2008; Breuer et al., 2009). Arable land is typically separated from the channel bank by a ca. 3-m-wide, near-channel vegetation zone of low-lying brush. Urban areas constitute nearly 10% of the basin area, primarily adjacent to the Dill River in the lower two-thirds of the basin (Breuer et al., 2009). Watershed characteristics are provided in Table 1.

Table 1
Characteristics of the Dill River basin.

Annual precipitation (mm) ^a	907.3
Annual precipitation range (mm) ^a	700–1300
Annual mean temperature (°C) ^a	8
Drainage area (km ²) ^b	692
Altitudinal range (m) ^b	150–700
Mean slope gradient (°) ^b	8
Maximum slope gradient (°) ^b	35
Mean annual discharge (mm) ^a	437.7
Land use in basin (% of basin area) ^a	
Cropland	6.5
Pasture	20.6
Fallow	9.1
Urban	9.2
Forest	54.3
Water	0.3

^a Breuer et al., 2009.

^b Fröhlich et al., 2008.

The floodplain is underlain by late Holocene (last 3000 years) alluvium and colluvium whose thickness in the adjacent Lahn and Wetter River basins ranges from 2.5–3 m in the Lahn watershed (Rittweger, 2000; Bos and Urz, 2003) to 1–2 m in the Wetter drainage basin (Houben, 2003). Research along the Lahn River suggests that floodplain alluvium accumulated during the late Holocene when the forest cover on surrounding uplands was removed for timber or cultivation, accelerating hillslope erosion and transporting sediment to the valley floor (Rittweger, 2000; Kalis et al., 2003; Hoffmann et al., 2007). Inceptisols are the most common USDA soil taxonomy type in the watershed, occupying nearly two-thirds of the catchment area (Breuer et al., 2009).

Potential sources of trace metals include former steel mills—where trace metals were used as alloys in steel manufacture in the upper part of the 25-km study reach near and upstream from Herborn (Fröhlich et al., 2008)—and the mines that were once active in uplands of the watershed (Meyer, 2007). Mining in the upper Dill River and lower Lahn River dates back many hundreds of years, but the peak period of mining was from 1870 to 1945; the last mine in the Dill and lower Lahn River basins, the Grube Fortuna iron mine, closed in 1983 (Meyer, 2007). Chiefly iron ore, but also some copper and lead ore, were mined in the Dill and lower Lahn River basins from Carboniferous shale; this same shale serves as a potential nonanthropogenic source of Cu, Pb, and Zn in the study reach (Wetzel, 1994). Industrial activity represents an additional possible source of trace metals in the basin. Through analysis of 73 streamwater sites along the entire 60 km length of the Dill River, Fröhlich et al. (2008) found maximum concentrations of trace metals in samples collected immediately downstream from steel mill locations. The reach where water samples contained the highest concentrations of trace metals corresponds to the 25-km reach of the Dill River examined in this study.

3. Methods

Through field reconnaissance, I selected 13 near-channel (<5 m from the active channel) sites along a 25-km reach of the Dill River between Herborn and Wetzlar. Sites were chosen at an interval of every 2–4 km along the river to ensure systematic coverage along the study reach while at that same time avoiding major roads, rail lines, and industrial facilities. Every attempt was made to select an equal number of left and right channel bank sites, although limited channel access made this difficult in a few reaches. The geographical latitude and longitude of each sample site was recorded using a hand-held GPS unit. Sample locations are shown on Fig. 1.

At each sample site, I collected a 2.5-cm diameter core from floodplain sediments using an Oakfield handheld corer to a depth of at least 1.0 m. In the field, cores were sampled at an interval of 5 cm for analysis of trace metals and standard sediment analyses. Samples were stored in sterile plastic bags and transported to the geomorphology lab at Justus-Liebig Universität in Giessen, where each sample was split, with half retained for grain size analysis and organic matter content determination and half submitted to ALS Chemex Labs (an ISO 902 registered analytical laboratory in North America) for analysis of trace metals. The trace metal samples were submitted to the ALS Chemex facility in Vancouver, BC, Canada, the same lab that I used for trace metal analysis on samples collected during prior research along the Lahn River. At the lab, trace metal analysis was completed on the <63-μm fraction. Samples were first digested in perchloric, nitric, and hydrofluoric acids (triple-acid digestion), a process that has been found to yield nearly total recovery of trace metals (Lecce and Pavlowsky, 1997). The concentrations of Cu, Pb, and Zn were measured using inductively coupled plasma atomic emission spectroscopy (ICE-AES). The analytical sensitivity was 1.0 mg kg⁻¹ for Cu and 2.0 mg kg⁻¹ for Pb and Zn. Quality control information provided by ALS Chemex yielded coefficients of variation on duplicate samples of 2.1% for Cu, 4.0% for Pb, and 1.4% for Zn. Repeated measurements of standards by the lab produced

coefficients of variation of 1.6% for Cu, 3.3% for Pb, and 1.1% for Zn. On the half of samples that I retained in the lab, the pipette method was used to determine the proportions of clay (<2 µm) and silt (2–62 µm); the percent sand was determined by wet sieving. Loss on ignition (LOI) at 430 °C for 24 h yielded the organic matter content of samples (Davies, 1974).

Samples collected at depths of 0.65 to 1.10 m were used to determine the background, or pre-industrial (pre-1750 A.D.), concentrations of trace metals. Concentrations in the background samples were then compared to concentrations in samples collected from the sediment cores to establish the degree of trace metal enrichment. As noted in previous research (e.g., Birch et al., 2000; Matschullat et al., 2000; Bindler et al., 2011), determining pre-industrial (sometimes referred to as nonanthropogenic) trace metal concentrations is problematic. Matschullat et al. (2000) summarized and assessed the options for determining background trace metal concentrations, including the use of global trace metal data as opposed to locally or regionally obtained background values. As I have done in previous work along the Lahn River (Martin, 2000, 2012), deeply buried sediments were collected and analyzed for trace metal concentrations to yield background values. Based on vertical metal trends and AMS radiocarbon ages on charcoal from sediment cores, sediments at depths >0.6 m date to the pre- or early-industrial period (early nineteenth century and older), and therefore are assumed to provide an accurate indication of background trace metal concentrations in the study area. In total, 19 samples provided background concentrations of Cu, Pb, and Zn.

To assess the degree of contamination, trace metal concentrations from sediment cores were compared to standards contained in the 1998 German Federal Soil Protection Act (*Bundes-Bodenschutzgesetz* {*BbodSchG*}) and the German Federal Soil Protection and Contaminated Site Ordinance of 1999 (*Bundes-Bodenschutzverordnung* {*BbodSchV*}). These federal acts spell out the trace metal concentration above which the 'natural soil functions' are thought to be at risk (the 'precautionary value'). Values are reported for loam/silt soils and are given in mg kg⁻¹ dry weight fine soil. I used threshold effect concentrations (TEC) and probable effect concentrations (PEC) outlined by MacDonald et al. (2000) as an additional measure to assess the degree of trace metal contamination. At concentrations less than the TEC, adverse impacts are unlikely; concentrations above the PEC indicate that negative impacts are likely.

Age control on sediment cores was afforded by radiocarbon ages on charcoal fragments recovered at several sample sites. In 8 of the 13 sediment cores, charcoal was recovered at depths ranging from 0.49 to 1.10 m. Charcoal was stored in sterile plastic bags and first refrigerated in the geomorphology lab at Giessen and later in the Paleoenvironmental lab of the Department of Geography at Kansas State University. Charcoal was submitted to Beta Analytic, Inc. for standard AMS radiocarbon dating. Dates on the charcoal provide age control on alluvial stratigraphy and an independent verification of the timing of trace metal accumulation in floodplain sediments.

Although regarded as suitable for radiocarbon dating (Pessenda et al., 2001; Hajdas et al., 2007; Higham et al., 2009), charcoal has potential limitations when used to date alluvial stratigraphy. Bioturbation can result in secondary mixing of charcoal and other organic materials (Adovasio et al., 1990; Pessenda et al., 2001); charcoal can be contaminated with secondary carbonate and organic acids, producing anomalous ages (Alon et al., 2002; Hajdas et al., 2007); and charcoal can be eroded from older sediments, thereby yielding an anomalously old age for sedimentation (Kennett et al., 2002; Seidel and Mäkel, 2007; Hoffmann et al., 2008). To avoid dating reworked charcoal, I used only charcoal fragments that appeared in large concentrations rather than as isolated fragments, given that the latter have a higher potential for being reworked from older deposits (DeLong and Arnold, 2007).

4. Results

4.1. Mean trace metal concentrations

The descriptive statistics for core samples and background samples are summarized in Table 2. Mean concentrations of Cu, Pb, and Zn from the core samples exceed the mean background concentrations. For the surface samples (upper 0.40 m), which represent sediments deposited after the onset of mining or industrial activity, a clear majority (97% for Cu, 94% for Pb, and 95% for Zn) have concentrations that are greater than the background concentrations. Results from a Mann–Whitney *U*-Test for unpaired sample means indicate that the mean concentrations of Cu, Pb, and Zn for samples from the upper 0.40 m of the cores (where trace metal concentrations are highest) are significantly greater ($p = 0.05$) than the mean concentrations for the background samples. For all samples, the mean concentrations of Cu, Pb, and Zn exceed the precautionary value contained in the German Federal Soil Protection Act and the TEC (threshold effect concentration), but are less than the PEC (probable effect concentration) of MacDonald et al. (2000). For the surface samples, mean concentrations of Cu, Pb, and Zn exceed the precautionary value and TEC value, but are less than the PEC value. For Cu, a clear majority of the surface samples exceed the precautionary value (96%) and TEC value (98%); only 7% of the surface samples have Cu concentrations that are greater than the PEC value. For Pb, 82% of the surface samples exceed the precautionary value and 97% are greater than the TEC value, but only 36% of the surface samples exceed the PEC value. For Zn, 92% of the surface samples have concentrations greater than the precautionary value, 93% have concentrations that are higher than the TEC, but only 29% exceed the PEC value. As indicated by the range of Cu, Pb, and Zn values, some individual sample sites have concentrations that far exceed the precautionary value, PEC, and TEC.

The mean concentration (upper 0.40 m) and maximum concentration of Cu, Pb, and Zn for each of the 13 sediment cores with distance

Table 2
Summary trace metal concentrations along the Dill River.^a

	Cu (mg kg ⁻¹)	Pb (mg kg ⁻¹)	Zn (mg kg ⁻¹)
<i>All (n = 213)</i>			
Mean	83.4	107.9	323.9
Median	80.0	99.0	345
SD	41.1	73.4	179.8
Range	25–229	20–408	73–992
<i>Surface samples (n = 118)</i>			
Mean	94.7	121.7	391.4
Median	88.5	114.0	391.0
SD	33.8	65.4	149.9
Range	28–229	25–408	79–992
<i>Subsurface samples (n = 95)</i>			
Mean	69.3	90.7	240.2
Median	52.0	55.0	153.0
SD	45.0	79.4	179.2
Range	25–219	20–355	73–671
<i>Background samples (n = 19)</i>			
Mean	37.5	39.3	105.0
Median	35.0	34.0	99.0
SD	9.9	17.6	30.1
Range	25–57	20–77	73–169
<i>Precautionary</i>			
Value	40.0	70.0	150.0
TEC	31.6	35.8	121.0
PEC	149.0	128.0	459.0

^a Surface samples are 0–0.40 m, subsurface samples are >0.40 m. Surface sample mean concentrations shown in bold are statistically different ($p = 0.05$) than background concentrations. The 'precautionary value' is published in the German Federal Soil Protection Act (*Bundes-Bodenschutzgesetz*) of 1998; the TEC (threshold effect concentration) and PEC (probable effect concentration) values are from MacDonald et al. (2000); all values are for dry weight sediment.

downstream along the study reach of the Dill River are shown in Fig. 2. Contrary to previous research, which frequently notes a downstream distance decay in trace metal concentrations (e.g., Bubb and Lester, 1996; Leigh, 1997; Miller, 1997; Lecce and Pavlowsky, 2014), the concentrations along the study reach generally do not decrease with distance downstream, with the exception of Pb whose mean and maximum concentrations decline slightly with distance downstream. As the r^2 and p values in Fig. 2 attest, the statistical relationship between distance downstream and trace metal concentrations is very weak, at best. Moving downstream, concentrations of Cu, Pb, and Zn vary in a nonsystematic fashion, with high mean and maximum concentrations present throughout the study reach. The absence of a decline in trace metals with distance downstream may be attributable to several non-point sources rather than a single point source for contaminants (Lecce and Pavlowsky, 2014).

4.2. Vertical trace metal trends

The vertical trends in trace metals, percent organic matter, and percent silt and clay for the eight sample cores that have AMS ^{14}C radiocarbon ages are shown in Fig. 3 (A–H); vertical trends in trace metals, percent organic matter, and percent silt and clay for the five sites that lack ^{14}C ages are similar to those depicted in Fig. 3. At the sites depicted, ^{14}C ages in the lower half of the cores provide chronological control on stratigraphy (see Kasielke and Zepp, 2011). The AMS dates, which were obtained on charcoal fragments, are provided in Table 3. A comparison of the percent silt and clay vertical trends with the trace metal vertical trends (Horowitz, 1985) reveals no relationship between sediment grain size and trace metal concentrations, similar to what I found in

previous work along the Lahn River (Martin, 2004). Likewise, organic matter content has minimal effect on trace element concentrations in the cores.

Generally, vertical trace metal peaks at individual core sites are found at similar depths, suggesting that post-depositional movement of metals has been minimal (Lecce and Pavlowsky, 2014). In a few cores, the peak concentrations for the metals occur at slightly different depths, usually within 0.10 m of one another. The different depths for peak concentrations may result from metals having various upstream sources that were active during different time periods, or result from post-depositional, down-core movement of metals. In those cores where the depth to peak concentrations differ, the metal trends are similar, an indication that any post-depositional migration of metals has been limited. The similarity in the depths to metals in individual cores and similar metal trends suggest that the physical deposition of metals by overbank flow rather than post-depositional migration of them accounts for metal trends, and they can thus be used as stratigraphic markers in cores. Typically cores extend to a depth of 0.90 to 1.0 m, but in some instances the presence of sandy or gravelly layers prevented deeper coring.

Vertical trends in trace metals are similar among cores, with a pronounced peak in concentrations at a specific depth below the floodplain surface. Often, the increase in metal concentrations to the peak occurs over a very short up-core distance (Fig. 3F). The maximum concentrations for Cu, Pb, and Zn are typically at depths ranging from 0.20 to 0.70 m. The mean depths of the peak concentrations of Cu, Pb, and Zn are 0.42, 0.38, and 0.36 m, respectively. For some metals in the three upstream cores (Fig. 3A, B, C), peak concentrations occur at two depths. For example in Fig. 3A, Zn concentrations peak at depths of 0.3 and 0.6 m. The presence of a double peak, especially evident in Fig. 3A for Zn and in Fig. 3B for Cu and Zn, suggests an upstream source that was present at a later date for those two trace metals. Detailed information on mining operations and industrial activity in the Dill River valley is unfortunately not available, so the secondary peaks in Cu and Zn cannot be linked to a particular source or sources of trace metals. It is possible that the secondary peaks are the result of remobilization by erosion of metal-rich sediments from upstream, near-channel locations and conveyance of them downstream. If that is the case, metal-rich sediments were deposited recently at the most upstream reaches of the study area because those sites are closest to the source or sources. Downstream sites lack the more recent secondary peaks because insufficient time has elapsed for those metal-contaminated sediments to be transported downstream. Interestingly, the secondary peak in Cu is present only in Fig. 3B. It may result from a recent local source of Cu immediately upstream from the site. In contrast, a pronounced double peak in Zn occurs in Fig. 3A and 3B, suggesting a more extensive upstream source (e.g., mine, industrial facility, or metals stored in near-channel sediments) for Zn.

Radiocarbon ages on charcoal range from 1670 ± 30 RCYBP (Beta-365179) to 90 ± 40 RCYBP (Beta-285118), indicating a late Holocene age for floodplain sediments. With the exception of cores D5 and D11 (Fig. 3D and 3F), the dated charcoal was recovered below the peak concentrations; at the aforementioned cores, the charcoal was recovered at nearly the same depth as peak trace metal concentrations. Radiocarbon ages generally fall within the time period mid/late-1700s to around 1900 and provide a maximum age for overlying sediments. Assuming that the maximum trace metal concentrations correspond to the period of peak iron ore mining and steel mill operations upstream (mid-1800s to mid-1900s), the AMS ages provide independent verification that deposition of metal-contaminated sediments commenced after the mid/late seventeenth century. Sediments deposited most recently (generally at depths above 0.20 m) are less contaminated with trace metals, an indication that cleaner sediments have begun to move through the watershed during the last several decades. A similar trend has been noted in other studies in western Europe that examined

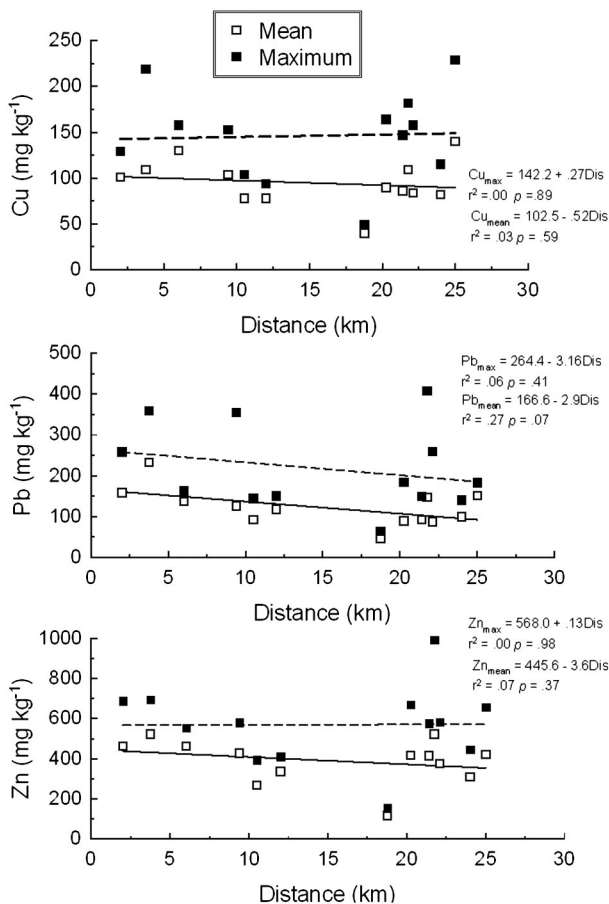


Fig. 2. Downstream trends in mean and maximum Cu, Pb, and Zn concentrations. The mean concentration was calculated from the upper 0.40 m of each core; the maximum concentration is the highest concentration for each trace metal from each core.

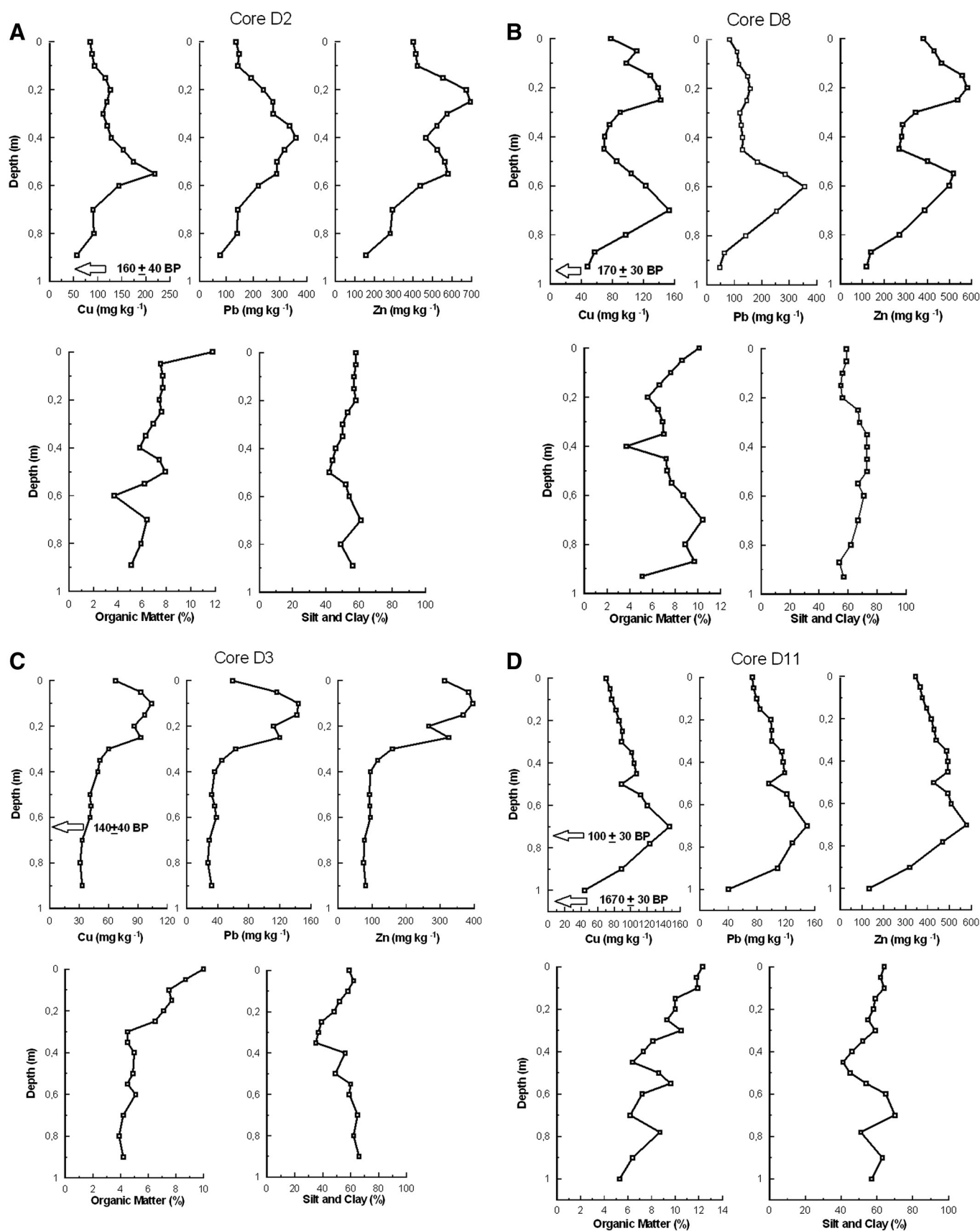


Fig. 3. Vertical trends in Cu, Pb, and Zn, organic matter content, and percentage silt and clay for the radiocarbon-dated cores. Site (A) is the most upstream location, site (H) is the most downstream location along the study reach.

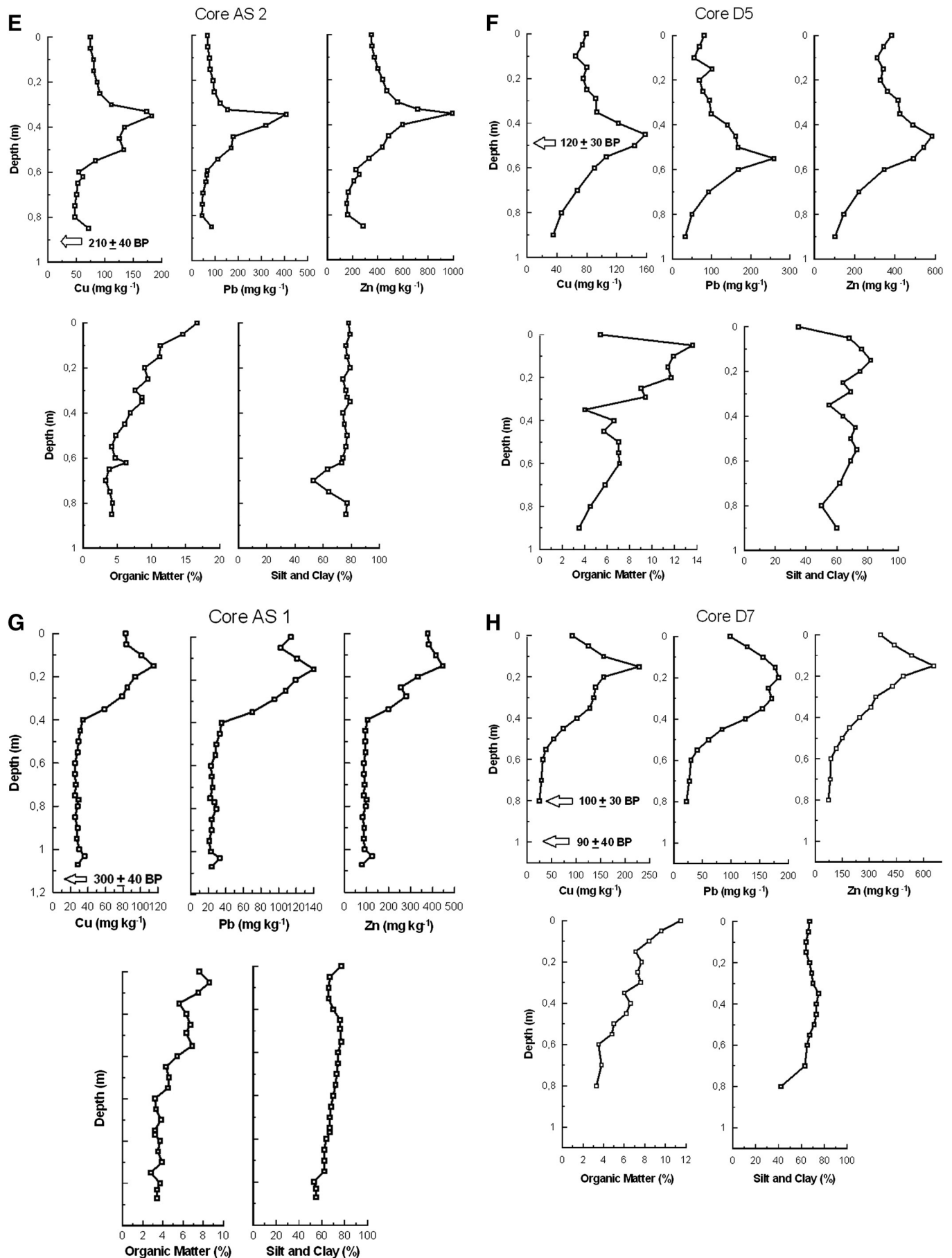


Fig. 3 (continued).

Table 3Results of AMS radiocarbon dating of floodplain sediments in the study reach.^a

Site	Lab. code	¹⁴ C-age (B.P.)	Cal. age	Position
D11-3	Beta-365180	100 ± 30	A.D. 1680–1950	Floodplain, 0.75 m depth
D11	Beta-365179	1670 ± 30	A.D. 262–426	Floodplain, 1.05 m depth
D8-13	Beta-365178	170 ± 30	A.D. 1660–1950	Floodplain, 0.95 m depth
D7-13	Beta-365177	100 ± 30	A.D. 1680–1950	Floodplain, 0.80 m depth
D5-2	Beta-331988	120 ± 30	A.D. 1670–1950	Floodplain, 0.49 m depth
D7	Beta-285118	90 ± 40	A.D. 1770–1950*	Floodplain, 1.05 m depth
D3	Beta-285117	140 ± 40	A.D. 1660–1960*	Floodplain, 0.63 m depth
D2	Beta-285116	160 ± 40	A.D. 1660–1960*	Floodplain, 0.90 m depth
AS2	Beta-201179	210 ± 40	A.D. 1640–1950*	Floodplain, 0.90 m depth
AS1	Beta-201178	300 ± 40	A.D. 1480–1660*	Floodplain, 1.10 m depth

^a Ages are from charcoal. Ages marked with an * are calibrated using the INTCAL04 database (Reimer et al., 2004); other ages are calibrated using the INTCAL09 database (Heaton et al., 2009; Reimer et al., 2009).

Table 4

Summary statistics for sedimentation rates of drainage basins in the vicinity of the Dill River.

Location	Sedimentation rate (cm y ⁻¹)	Total sediment thickness (m)	Time period (YBP)	Reference
Amöneburg Basin, Germany	0.01–0.06	0.5–2.0	3500–4000	Rittweger (2000), Kalis et al. (2003)
Lahn River, Germany	0.07	2.5	3500–4000	Lang and Nolte (1999)
Aufsess River, Germany	0.1	1.0	1000	Fuchs et al. (2011)
Vills River, Germany	0.28–0.35	2.0–2.5	700	Raab et al. (2011)
Kalwes River, Germany	0.4	2.0	500	Kasielke and Zepp (2011)
Dill River, Germany	0.2–0.3	0.6–1.0	150–300	This study

vertical trends of trace metals in recent floodplain sediments (Brüggemann, 1995; Martin, 1997, 2000; Zerling et al., 2006; Zachmann et al., 2013).

In a downstream direction, the depth of Cu, Pb, and Zn peak concentrations varies from site to site. As an example, at core D2 (Fig. 3A), an upstream site, peak concentrations of Cu, Pb, and Zn occur at depths of 0.5, 0.4, and 0.3 m, respectively; in contrast, at core D8 (Fig. 3B), which was collected 7.5 km farther downstream, the peak concentrations of Cu, Pb, and Zn are present at 0.7, 0.6, and 0.6 m, respectively. The most downstream sample (core D7 at 25 km downstream; Fig. 3H) contains maximum concentrations of all three trace elements at 0.2 m. The absence of a systematic downstream trend in the depth to maximum trace metal concentrations may be linked to geomorphic variables such as floodplain elevation above the active river channel (Lecce and Pavlowsky, 2014). Multiple point sources of trace metals downstream (e.g., industrial facilities) may also affect the depth at which maximum metal concentrations are found by supplying high loads of contaminants to the river, thereby masking the contribution of upstream metal sources. This may explain the shallow depth at which peak concentrations are found in core D7 (Fig. 3H), which is located just downstream from several industrial facilities. The different depths to peak concentrations within individual cores, as noted previously, may result from metals having various upstream sources that were active during different time periods or be the product of some post-depositional, down-core movement of metals.

4.3. Recent floodplain sedimentation

Radiocarbon ages from 8 of the 13 sediment cores allow estimation of the amount of recent floodplain sedimentation in the study reach. Estimates of late Holocene floodplain sedimentation from this study and from drainage basins in the vicinity of the Dill River are summarized in Table 4. The radiocarbon ages from the Dill River floodplain indicate 0.6 to 1 m of sedimentation over the last 150–300 years, or a sedimentation rate of 0.2–0.3 cm y⁻¹; the amount and rate of deposition are greater than those found in other watersheds in the vicinity of the study reach. Core D7 (Fig. 3H), the most downstream sample location, had the greatest sedimentation of 0.8 to 1 m over about a century (0.8 cm y⁻¹); in contrast, at core AS1 (Fig. 3G), which is 1 km

upstream of site D7, 1.1 m of sedimentation over 300 years occurred (0.4 cm y⁻¹). The other sites with radiocarbon control suggest between 0.6 and 0.8 m of sedimentation over 150 to 200 years (0.3–0.4 cm y⁻¹). No downstream trend in the sedimentation amount was apparent; some downstream sites (D7; Fig. 3H) had high sedimentation, whereas others (AS1; Fig. 3G) had much less.

4.4. Trace metal storage in the Dill and Lahn river valleys

The mean trace metal concentrations noted here along the Dill River exceed the values that I found at near-channel positions along the Lahn River (Martin, 2004, 2012). Mean trace metal concentrations from 28 near-channel sites in the Marburg to Wetzlar reach of the Lahn River, which is upstream of the Lahn River confluence with the Dill River at Wetzlar, are summarized in Table 5 as are mean trace metal concentrations at depths of 0–0.05 m along the Dill River. As the data reveal, mean concentrations along the Dill River are higher than mean concentrations along the Lahn River. A Mann–Whitney *U*-Test for unpaired sample means indicates that the Dill River mean concentrations of all three trace elements are significantly higher than mean concentrations from the Lahn River samples ($p < 0.01$). The findings suggest that floodplain

Table 5Summary trace metal concentrations along the Dill River and Lahn River.^a

	Cu (mg kg ⁻¹)	Pb (mg kg ⁻¹)	Zn (mg kg ⁻¹)
<i>Dill River (n = 26)</i>			
Mean	80.9	92.7	354.6
Median	74.5	89.0	361.5
SD	21.5	28.3	75.5
Range	41–141	57–150	147–505
<i>Lahn River (n = 28)</i>			
Mean	39.1	46.8	233.6
Median	37.0	44.0	204.5
SD	8.2	9.6	57.4
Range	27–56	34–64	146–343

^a Samples at both locations collected at depths of 0–0.05 m. Dill River surface sample mean concentrations shown in bold are statistically different ($p < 0.01$) than the mean concentrations from Lahn River samples.

sediments along the Dill River contain higher concentrations of Cu, Pb, and Zn than do sediments in a similar geomorphic position along the Lahn River upstream of the confluence of the Lahn and Dill Rivers at Wetzlar. The source of these additional trace metals is likely anthropogenic activity in the form of mining, industrial activity, or both.

5. Discussion

Core samples collected at 13 near-channel sites along the floodplain of the Dill River reveal concentrations of Cu, Pb, and Zn that exceed background concentrations, with the highest level of trace metals in the upper 0.40 m of cores. Mean concentrations of Cu, Pb, and Zn exceed the 'precautionary value' contained in the German Federal Soil Protection Act (*Bundes-Bodenschutzgesetz* {*BodSchG*}); some individual samples have concentrations of Cu, Pb, and Zn that are several times the 'precautionary value'. Nearly all samples in the upper 0.4 m contain concentrations of Cu, Pb, and Zn that exceed the 'precautionary value' of the German standards and the TEC (threshold effect contamination) of McDonald et al. (2000). The peak concentration of all three trace metals tend to be found at depths of 0.20 to 0.70 m, with lower concentrations up-core toward the surface. This trend suggests that the release of the trace metals to the environment has decreased over the last several decades, a finding noted in other studies from western Europe of metal contamination levels in floodplain sediments (e.g., Brüggmann, 1995; Scherer et al., 2003; Martin, 2012; Zachmann et al., 2013; Vallius, 2014).

Mean and maximum trace metal concentration do not systematically decrease with distance downstream (Fig. 2), suggesting multiple nonpoint sources rather than a single point source for trace metal contamination in the watershed (Lecce and Pavlowsky, 2014). Although the upper Dill River basin has a history of iron ore mining, with the peak period of mining activity between 1870 and 1945, there are other possible sources for trace metals in the catchment, including but not limited to industrial activity in upstream urban areas such as Dillenburg and Herborn. A variety of sources for the trace metals likely account for the absence of a downstream distance decay in metal contamination. Given multiple sources for metals along the river, a decrease in metal concentrations with distance downstream would be unexpected as additional metals have been introduced to the study reach at various locations. The absence of a distance decay downstream could also reflect the episodic movement of metals downstream through the study reach. Although reductions in the release of trace metals to watersheds have been linked to reduced contamination in downstream floodplain sediments (Ciszewski, 2003; Grosbois et al., 2006; Bábek et al., 2008), several decades may pass before declines in metal releases are reflected in downstream sediments (Outridge et al., 2011). If that is the case, near-channel sediments along the Dill River may serve as potential sources of downstream trace metal pollution in the foreseeable future, something noted by others working in western European watersheds (e.g., Förstner, 2003; Krüger et al., 2005; Macklin et al., 2006; Foulds et al., 2014; Resongles et al., 2014).

General knowledge of the upstream mining history in the Dill River and AMS radiocarbon ages on charcoal from the sediment cores provide chronologic control on recent floodplain sedimentation in the study reach. The vertical trends in trace metal concentrations display a pronounced peak at specific depths below the floodplain surface, generally between 0.20 and 0.70 m (Fig. 3, A–H). The peak is presumed to correspond with the period of maximum metal releases to the environment through mining and industrial activities (e.g., steel mill operations) upstream in the watershed. Similarities among vertical trends in cores indicate that post-depositional movement of metals has been minimal. The radiocarbon ages, which come from depths at or below the peak metal concentrations, provide maximum ages for sediments at depths of 0.60 to 1.10 m. The radiocarbon ages indicate that deposition of metal-contaminated sediments commenced after the

mid/late seventeenth century, which corresponds well with the history of metal releases to the environment. With the exception of core D7 (Fig. 3H), the other radiocarbon-dated cores place the peak metal concentration at 100 to 150 years ago, or during the late 1800s and early 1900s. Cores D2, D11, AS2, and D5 (Fig. 3A, D, E, F), which range from the upstream reach of the study area to the most downstream reach, all suggest the deposition of peak metal-contaminated sediments in the late 1800s to early 1900s. The sample site whose vertical metal trends and radiocarbon ages are an anomaly is core D7 (Fig. 3H). Here, the radiocarbon ages are very young, and the peak metal concentrations occur at an unusually shallow depth of 0.15–0.20 m. As the most downstream sample site, the location may have received additions of trace metals eroded from upstream sites since the mid-1900s.

Aside from site D7, where radiocarbon ages indicate about 1.0 m of sediment deposition during the 1900s, radiocarbon ages and trace metal trends from the other cores in the study reach document 0.6 to 1.1 m of floodplain sedimentation over the last 150 to 300 years ($0.2\text{--}0.3\text{ cm y}^{-1}$). This amount exceeds recent floodplain sedimentation documented in nearby watersheds (Lang and Nolte, 1999; Rittweger, 2000; Kalis et al., 2003; Houben, 2012). Work elsewhere in Germany has noted rapid increases in floodplain sedimentation beginning about 1000 years ago. Increased sedimentation has been ascribed to expanded agricultural activity, wetter climate conditions, or a combination thereof (Fuchs et al., 2011; Raab et al., 2011). There is little evidence in the literature of high rates of sedimentation such as those documented here along the Dill River having occurred over the last 300 years. The relatively rugged terrain of the Dill River basin may have resulted in high sediment delivery to the valley over the last several hundred years, especially during periods when the forest cover was reduced by expansion of agriculture and perhaps mining activity. The uncertainty involved with radiocarbon ages notwithstanding, the calculated sedimentation rates in the study reach range from $0.06\text{ cm to }1.2\text{ cm y}^{-1}$, with a mean calculated sedimentation rate for the dated cores ($n = 10$) of 0.56 cm y^{-1} (standard deviation = 0.3 cm y^{-1}), a rate higher than has been noted in surrounding watersheds.

Initially, I began work in the Dill River basin after noting that trace metal storage along the Lahn River was greater downstream from Wetzlar, which marks the confluence of the Lahn and Dill Rivers (Martin, 2004, 2012). Results here from the core samples along the Dill River indicate that the concentration of trace metals in floodplain sediments is higher in the Dill River valley than in the Lahn River valley (Table 5). These findings suggest that at least some of the store of metals downstream from Wetzlar in the Lahn valley can be attributed to metal-laden sediments being conveyed down the Dill River, although the exact origin of the metals (mining, industry, or other source) cannot be established. Trace metals stored in near-channel positions of the Dill River serve as a potential source for future contaminants in reaches of the Lahn River downstream from Wetzlar, potentially extending to the confluence of the Lahn and Rhine Rivers near Koblenz. Depending on the rate with which metals stored along the Dill River are remobilized, it could be decades to centuries before trace metal concentrations in floodplain sediments of the lower Lahn River approach the lower concentrations present along the upper Lahn River (Martin, 2009, 2012).

6. Conclusions

Mining and industrial activity in the upper reaches of the Dill River in central Germany have left a chemical imprint on downstream reaches of the river. Concentrations of Cu, Pb, and Zn in cores collected from floodplain sediments along a 25-km reach of the Dill River between Herborn and Wetzlar, Germany, exceed background concentrations, providing evidence of anthropogenic enrichment of trace metals in the Dill valley. In the upper 0.40 m of the cores, a clear majority of the samples (97% for Cu, 94% for Pb, 95% for Zn) had concentrations that were greater than the background

values. Mean concentrations of Cu, Pb, and Zn are greater than the 'precautionary value' contained in the German Federal Soil Protection Act (*Bundes-Bodenschutzgesetz* {*BodSchG*}) and the 'threshold effect concentration' (TEC) of MacDonald et al. (2000). Several individual samples have Cu, Pb, and Zn concentrations that are at least twice the TEC and PEC values. Together, results from the core samples document significant storage of anthropogenic Cu, Pb, and Zn in geomorphically sensitive near-channel positions along the Dill River. Should these metal-laden sediments be eroded in the future, they would serve as sources of trace metal pollution downstream along the Dill and Lahn Rivers. Such storage has been referred to as saturation (Nriagu et al., 1998; Outridge et al., 2011), and can result in a lag between reduced metal releases to the environment and a corresponding decrease in metal concentrations in downstream floodplain sediments. The absence of a distance decay in the mean and maximum concentrations of trace elements along the Dill River is linked to additional downstream sources that have delivered trace metals to the Dill River.

Vertical trace metal trends reveal a distinct peak in concentrations, typically at depths of 0.20 to 0.70 m. Trace metal trends are generally similar among sediment cores, an indication that the trends reflect mainly the deposition of the metal-laden sediments rather than post-depositional translocation of metals. The upper 0.20 m of most cores contains lower concentrations of trace metals, suggesting that cleaner sediments have been deposited during the past few decades. The peak metal concentrations are assumed to correspond to the maximum release of contaminants to the environment. Although it is not possible to determine the specific source of metals, the Dill River basin upstream from the study reach has a history of iron-ore mining along tributaries and steel processing in cities such as Herborn and Dillenburg (Fröhlich et al., 2008). The high point of such activity was the mid-1800s to the mid-1900s. The peak of trace metal concentrations is assumed to date to that period and is supported by AMS radiocarbon ages indicating that sediments at depths of 0.5 to 1.10 m were deposited between 90 and 1700 years ago. The radiocarbon ages, which were obtained from charcoal recovered at or below the depth of peak metal concentrations, provide a maximum age for the accumulation of metal-laden sediments. Radiocarbon ages and the depth to the peak concentration of metals indicate 0.6 to 1 m of floodplain sedimentation over the last 150–300 years ($0.2\text{--}0.3\text{ cm y}^{-1}$), a rate exceeding that noted in other studies in the vicinity of the Dill River (Lang and Nolte, 1999; Rittweger, 2000; Kalis et al., 2003; Houben, 2012). The relatively high sedimentation rate along the Dill may be the result of the high relief of the watershed, which would have favored delivery of high amounts of sediment to the river valley during periods of expanded agriculture or mining activity.

The Dill River appears to be contributing to the relatively high concentrations of Cu, Pb, and Zn measured in near-channel positions along the Lahn River downstream from Wetzlar, where the Dill and Lahn rivers join (Martin, 2009, 2012). Although Martin (2012) noted that surface sediments along the Lahn became less contaminated with trace metals between 2000 and 2007, contaminants stored along the Dill River have the potential of serving as a significant legacy source for metals in the foreseeable future. Channel bank erosion along the Dill River could entrain metal-contaminated sediments and convey them to the Lahn River, posing a threat to water and floodplain soil quality. Years after industrial sources were shuttered, emissions regulated, and mining activity terminated, trace metals stored in geomorphically sensitive positions along the Dill River represent potential sources of contamination for the foreseeable future.

Acknowledgments

This study was supported by a research fellowship from the Alexander von Humboldt-Stiftung of Bonn, Germany, and began during a stay as a research fellow in the Institut für Geographie at Justus Liebig-Universität in Giessen, Germany. I gratefully acknowledge the assistance of Professor Lorenz King, who served as my

host professor during the initial stay at the institute, and Professor Markus Fuchs, who supported me during subsequent visits to the institute and provided access to the geomorphology lab. Mr. Matthias Schick contributed valuable assistance during the laboratory work. I thank my colleague Professor Kendra McLauchlan for providing access to the Paleoenvironmental Lab at Kansas State University. The valuable comments and suggestions from two anonymous reviewers greatly improved the manuscript.

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